— *Questions marked with a star* ⋆ *are intended to be more difficult and may be skipped* —

1. Without computing any integrals, find the (complex) Fourier series of

$$\text{(i) } f(t) = \cos^3(2t), \quad \text{(ii) } g(t) = \cos(2t)\sin(4t),$$
$$\text{(iii) } h(t) = \sin(t)\cos(2t) + \sin(2t)\cos(t),$$

Using these results, find the trigonometric Fourier series of the above functions.

Hint. Recall that $\cos x = \frac{1}{2}(\mathrm{e}^{\mathrm{i}x} + \mathrm{e}^{-\mathrm{i}x})$ and $\sin x = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x})$.

Solution. (i) Expanding $f(t)$ into complex exponentials, we see that

$$\begin{aligned} f(t) &= \left(\frac{\mathrm{e}^{2\mathrm{i}t} + \mathrm{e}^{-2\mathrm{i}t}}{2}\right)^3, \\ &= \frac{1}{8}(\mathrm{e}^{6\mathrm{i}t} + 3\mathrm{e}^{2\mathrm{i}t} + 3\mathrm{e}^{-2\mathrm{i}t} + \mathrm{e}^{-6\mathrm{i}t}), \\ &= \tfrac{1}{4}\cos(6t) + \tfrac{3}{4}\cos(2t). \end{aligned}$$

(ii) Doing the same for $g(t)$ gives

$$\begin{aligned} g(t) &= \left(\frac{\mathrm{e}^{2\mathrm{i}t} + \mathrm{e}^{-2\mathrm{i}t}}{2}\right)\left(\frac{\mathrm{e}^{4\mathrm{i}t} - \mathrm{e}^{-4\mathrm{i}t}}{2\mathrm{i}}\right), \\ &= \frac{1}{4\mathrm{i}}(\mathrm{e}^{6\mathrm{i}t} + \mathrm{e}^{2\mathrm{i}t} - \mathrm{e}^{-2\mathrm{i}t} - \mathrm{e}^{-6\mathrm{i}t}), \\ &= \tfrac{1}{2}\sin(6t) + \tfrac{1}{2}\sin(2t). \end{aligned}$$

(iii) We could proceed similarly for $h(t)$, but this can be more easily done using the sine addition formula. Concretely, we have

$$h(t) = \sin(3t) = \frac{\mathrm{e}^{3\mathrm{i}t} - \mathrm{e}^{-3\mathrm{i}t}}{2\mathrm{i}}.$$

2. Find the complex Fourier series of the 2-periodic function defined on $[0, 2)$ by

$$f(t) = \begin{cases} t, & 0 \le t \le 1, \\ 1, & 1 < t < 2. \end{cases}$$

Solution. As f has period 2, the fundamental angular frequency Ω is π. The complex Fourier coefficients are therefore

$$c_n = \frac{1}{2}\int_0^2 f(t)\mathrm{e}^{-\mathrm{i}n\pi t}\,\mathrm{d}t = \frac{1}{2}\int_0^1 t\mathrm{e}^{-\mathrm{i}nt}\,\mathrm{d}t + \frac{1}{2}\int_1^2 \mathrm{e}^{-\mathrm{i}n\pi t}\,\mathrm{d}t.$$

For $n \neq 0$, this equals

$$\begin{aligned} c_n &= \frac{1}{2}\left[-\frac{te^{-in\pi t}}{in\pi}\right]_0^1 + \frac{1}{2in\pi}\int_0^1 e^{-in\pi t}\,dt - \frac{1}{2in\pi}\left[e^{-in\pi t}\right]_1^2 \\ &= \frac{(-1)^{n+1}}{2in\pi} - \frac{1}{2in\pi}\left(1+(-1)^{n+1}\right) - \frac{1}{2(in\pi)^2}\left[e^{-in\pi t}\right]_0^1 \\ &= -\frac{1}{2in\pi} + \frac{(-1)^n - 1}{2\pi^2 n^2}. \end{aligned}$$

Now taking $n = 0$, we see that c_0 is given by

$$c_0 = \frac{1}{2}\int_0^1 t\,dt + \frac{1}{2}\int_1^2 dt = \frac{3}{4}.$$

Therefore, the Fourier series is given by

$$f(t) \sim \frac{3}{4} + \sum_{n\neq 0}\left(-\frac{1}{2in\pi} + \frac{(-1)^n - 1}{2\pi^2 n^2}\right)e^{in\pi t}.$$

3. Let $f : \mathbb{R} \to \mathbb{C}$ be the 2π-periodic function defined by $f(t) = e^{|t|}$ on the interval $[-\pi, \pi)$. Show that the Fourier series of f is given by

$$f(t) \sim \frac{1}{\pi}\sum_{n=-\infty}^{\infty} \frac{(-1)^n e^{\pi} - 1}{n^2+1} e^{int}.$$

Without computing any further integrals, compute the trigonometric form of the Fourier series.

Solution. The function $f(t)$ equals e^t when $t > 0$ and e^{-t} when $t < 0$, so we split up the integral for c_n as

$$c_n = \frac{1}{2\pi}\int_{-\pi}^0 e^{-t-int}\,dt + \frac{1}{2\pi}\int_0^{\pi} e^{t-int}\,dt.$$

Integrating these we get

$$\begin{aligned} c_n &= \frac{1}{2\pi}\left[-\frac{e^{-t-int}}{1+in}\right]_{-\pi}^0 + \frac{1}{2\pi}\left[\frac{e^{t-int}}{1-in}\right]_0^{\pi} \\ &= \frac{1}{2\pi(1+in)}(e^{\pi+in\pi} - 1) + \frac{1}{2\pi(1-in)}(e^{\pi-in\pi} - 1). \end{aligned}$$

Using $e^{in\pi} = e^{-in\pi} = (-1)^n$, we can simplify this to

$$c_n = \frac{(-1)^n e^{\pi} - 1}{2\pi}\left(\frac{1}{1+in} + \frac{1}{1-in}\right) = \frac{(-1)^n e^{\pi} - 1}{\pi(n^2+1)},$$

giving the required series. Note that $f(t)$ is even, so the sine coefficients b_n vanish, while

$$a_n = c_n + c_{-n} = 2c_n = \frac{2}{\pi}\frac{(-1)^n \mathrm{e}^{\pi} - 1}{n^2 + 1}.$$

Therefore, the trigonometric series is

$$f(t) \sim \frac{\mathrm{e}^{\pi} - 1}{\pi} + \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{(-1)^n \mathrm{e}^{\pi} - 1}{n^2 + 1} \cos nt.$$

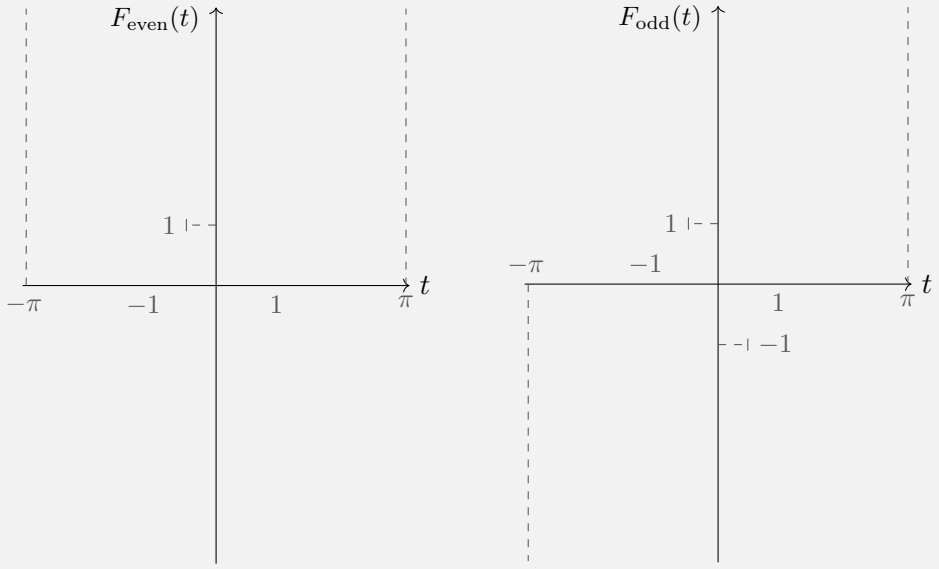
4. Plot the even and odd extensions of the following functions:

$$\begin{aligned} &\text{(i)} \quad f(t) = (t-1)^2, && 0 < t < \pi, \\ &\text{(ii)} \quad g(t) = \mathrm{e}^t - 1, && 0 < t < 1, \\ &\text{(iii)} \quad h(t) = \frac{1}{1+t}, && 0 < t < 2. \end{aligned}$$

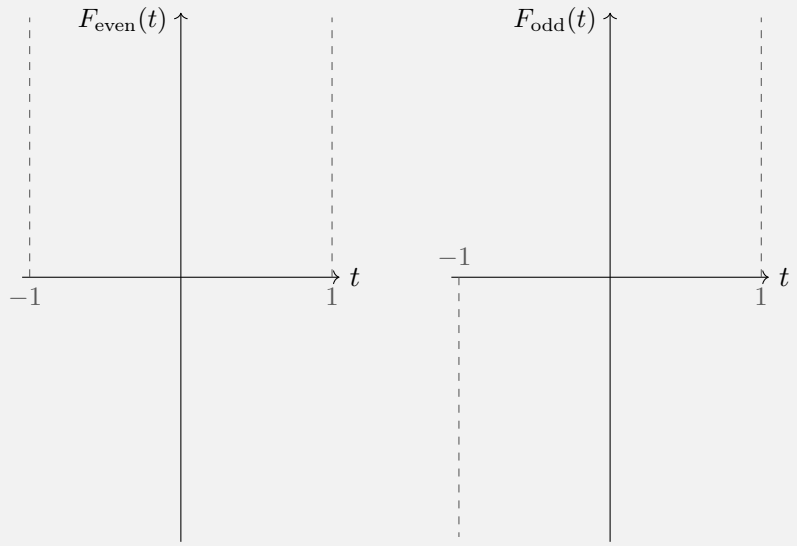
In each case indicate where the function meets the coordinate axes. Find the cosine half-range series of $f(t)$.

Solution. In what follows the original function is plotted in blue, while the even extension is in red and the odd extension is in purple for $t < 0$.

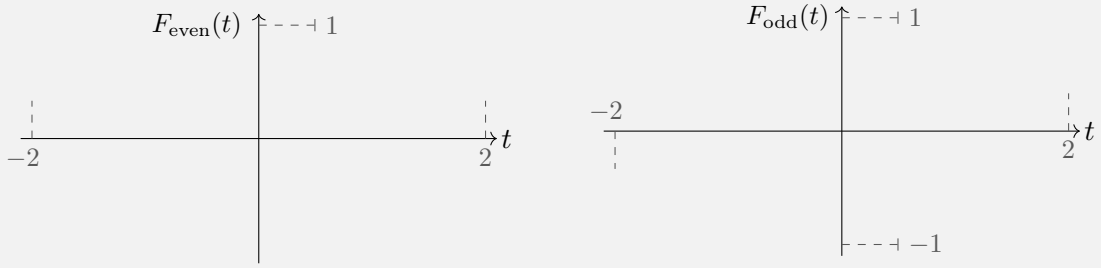
(i)



(ii)



(iii)



As for the cosine half-range series of $f(t)$, the coefficients can be found by integration by parts, resulting in

$$f(t) \sim \frac{\pi^2 - 3\pi + 3}{3} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(\pi - 1)(-1)^n + 1}{n^2} \cos nt.$$

5. Let $f : [0, 2) \to \mathbb{R}$ be the function

$$f(t) = \begin{cases} t, & 0 \leq t < 1, \\ 2 - t, & 1 \leq t < 2. \end{cases}$$

Show that the sine half-range series of f is given by

$$f(t) \sim \sum_{n=1}^{\infty} \frac{8}{\pi^2 n^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{\pi n t}{2}\right).$$

Solution. We extend $f(t)$ to an odd function on $(-2, 2)$ and then extend periodically. Therefore, $T = 4$ and $\Omega = \pi/2$. Hence

$$b_n = \int_0^1 t \sin\left(\frac{\pi n t}{2}\right) \mathrm{d}t + \int_1^2 (2-t) \sin\left(\frac{\pi n t}{2}\right) \mathrm{d}t.$$

Integration by parts gives

$$\begin{aligned} b_n &= -\frac{2}{\pi n}\left[t \cos\left(\frac{\pi n t}{2}\right)\right]_0^1 + \frac{2}{\pi n}\int_0^1 \cos\left(\frac{\pi n t}{2}\right) \mathrm{d}t \\ &\quad - \frac{2}{\pi n}\left[(2-t)\cos\left(\frac{\pi n t}{2}\right)\right]_1^2 - \frac{2}{\pi n}\int_1^2 \cos\left(\frac{\pi n t}{2}\right) \mathrm{d}t, \\ &= -\frac{2\cos\left(\frac{\pi n}{2}\right)}{\pi n} + \frac{4}{(\pi n)^2}\left[\sin\left(\frac{\pi n t}{2}\right)\right]_0^1 \\ &\quad + \frac{2\cos\left(\frac{\pi n}{2}\right)}{\pi n} - \frac{4}{(\pi n)^2}\left[\sin\left(\frac{\pi n t}{2}\right)\right]_1^2. \end{aligned}$$

The two unbracketed terms cancel, while only the $t = 1$ boundary terms are non-zero, leading to

$$b_n = \frac{8}{\pi^2 n^2} \sin\left(\frac{\pi n}{2}\right)$$

Therefore, the sine series for $f(t)$ is

$$f(t) \sim \sum_{n=1}^{\infty} \frac{8}{\pi^2 n^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{\pi n t}{2}\right).$$

We can simplify this slightly by noting that $\sin(n\pi/2) = 0$ when n is even. As such, we can define $n = 2k - 1$ to get

$$f(t) \sim \sum_{k=1} \frac{8(-1)^{k+1}}{\pi^2 (2k-1)^2} \sin\left(\frac{(2k-1)\pi t}{2}\right).$$

6. (⋆) Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by $f(t) = \sqrt{t}$ on $[0, \pi)$. Show that the sine half-range series can be written as

$$f(t) \sim \sum_{n=1}^{\infty} \left(\frac{2(-1)^{n+1}}{n\sqrt{\pi}} + \frac{2C(\sqrt{\pi n})}{\pi n^{3/2}}\right) \sin nt,$$

where $C(x)$ is the Fresnel integral

$$C(x) = \int_0^x \cos(t^2) \, \mathrm{d}t.$$

Solution. Taking the odd extension we get the period $T = 2\pi$, so $\Omega = 1$. The sine coefficients are given by

$$b_n = \frac{2}{\pi}\int_0^\pi \sqrt{t}\sin(nt)\,\mathrm{d}t,$$

for $n \geq 1$. To get rid of the square root, we make the substitution $u = \sqrt{t}$, giving the measure $\mathrm{d}t = 2u\,\mathrm{d}u$ and changing the integration to $(0, \sqrt{\pi})$. Therefore,

$$b_n = \frac{4}{\pi}\int_0^{\sqrt{\pi}} u^2\sin(nu^2)\,\mathrm{d}u.$$

We can integrate this by parts by noting that

$$-\frac{1}{2n}\frac{\mathrm{d}}{\mathrm{d}u}\left(\cos(nu^2)\right) = u\sin(nu^2),$$

hence

$$b_n = \left[-\frac{2u\cos(nu^2)}{\pi n}\right]_0^{\sqrt{\pi}} + \frac{2}{\pi n}\int_0^{\sqrt{\pi}}\cos(nu^2)\,\mathrm{d}u.$$

Substituting the boundaries into the first term gives $2(-1)^{n+1}/n\sqrt{\pi}$, while we can make the substitution $v = \sqrt{n}u$ in the second term to arrive at

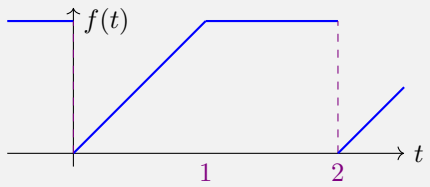
$$\begin{aligned} b_n &= \frac{2(-1)^{n+1}}{n\sqrt{\pi}} + \frac{2}{\pi n^{3/2}}\int_0^{\sqrt{\pi n}}\cos(v^2)\,\mathrm{d}v, \\ &= \frac{2(-1)^{n+1}}{n\sqrt{\pi}} + \frac{2C(\sqrt{\pi n})}{\pi n^{3/2}}, \end{aligned}$$

giving the result.

7. Show that the function $f(t)$ given in problem 2 satisfies the Dirichlet criterion. By evaluating the Fourier series of f at $t = 0$ and equating it to the appropriate value, derive that

$$\sum_{n=1}^{\infty}\frac{1+(-1)^{n+1}}{n^2} = \frac{\pi^2}{4}.$$

Solution. Before proceeding, it is useful to plot the function.



To establish that f satisfies the Dirichlet criterion, we must show that the left and right derivatives $f'_{\pm}(t)$ exist everywhere. Inspecting the graph, we see that away from $t = 0$ and $t = 1$, the function is differentiable, so $f'_-(t) = f'_+(t)$ exists there. At $t = 0$ and $t = 1$, the function fails to be differentiable and we must calculate $f'_{\pm}(t)$ explicitly.

First consider $t = 0$. Here the left limit $f(0^-)$ is 1, while the right limit $f(0^+)$ is 0. We can figure out the left and right derivatives by inspection, but we will calculate them explicitly. The left derivative is

$$f'_-(0) = \lim_{h\to 0^-} \frac{f(h) - f(0^-)}{h} = \lim_{h\to 0^-} \frac{1-1}{h} = 0.$$

Here we're using the fact that $h < 0$, so $f(h) = 1$ as $h \to 0^-$. Similarly, the right derivative is

$$f'_+(0) = \lim_{h\to 0^+} \frac{f(h) - f(0^+)}{h} = \lim_{h\to 0^+} \frac{h-0}{h} = 1.$$

Again, we're remembering that $h > 0$, so $f(h) = h$ as $h \to 0^+$.

Turning to $t = 1$, the function is continuous, so the left and right limits are both equal to $f(1^-) = f(1^+) = 1$. The left derivative is therefore

$$f'_-(1) = \lim_{h\to 0^-} \frac{f(1+h) - f(1^-)}{h} = \lim_{h\to 0^-} \frac{(1+h)-1}{h} = 1,$$

while the right derivative is

$$f'_+(1) = \lim_{h\to 0^+} \frac{f(1+h) - f(1^+)}{h} = \lim_{h\to 0^+} \frac{1-1}{h} = 0.$$

Therefore, the left and right derivatives exist everywhere and f satisfies the Dirichlet criterion.

Now that we know f satisfies the Dirichlet criterion, we know that the Fourier series at $t = 0$ equals the average of the left and right limits of f. Therefore

$$\frac{1}{2} = \frac{3}{4} + \sum_{n\neq 0} \left(-\frac{1}{2in\pi} + \frac{(-1)^n - 1}{2\pi^2 n^2} \right).$$

Currently, the sum runs over all of $\mathbb{Z}$ (except for $n = 0$), but we'd like it to run from $n = 1$ to ∞. We do this by pairing up the terms corresponding to n and $-n$. Doing so cancels out the first term in the bracket and we are left with

$$-\frac{1}{4} = \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{\pi^2 n^2}.$$

Multiplying by $-\pi^2$, we arrive at the result

$$\sum_{n=1}^{\infty} \frac{1+(-1)^{n+1}}{n^2} = \frac{\pi^2}{4}.$$

8. By applying Parseval's theorem to the sine half-range series for the function $f(t)$ in problem 5, show that

$$\sum_{k=1}^{\infty} \frac{1}{(2k-1)^4} = \frac{\pi^4}{96}.$$

Hint. Remember that the sine half-range series is the Fourier series of the odd extension F_{odd} of f.

Solution. As shown in problem 5, the sine half-range coefficients of $f(t)$ are given by

$$b_n = \frac{8}{\pi^2 n^2} \sin\left(\frac{\pi n}{2}\right).$$

Since $\sin(\pi n/2) = 0$ when n is even, we let $n = 2k-1$ and get

$$b_{2k-1} = \frac{8(-1)^{k+1}}{\pi^2 n^2}.$$

Applying Parseval's theorem to the odd extension of f gives

$$\frac{1}{4}\int_{-2}^{2} |F_{\text{odd}}((t)|^2 \, \mathrm{d}t = \frac{1}{2}\sum_{k=1}^{\infty} \frac{64}{\pi^4(2k-1)^4}.$$

The function $|F_{\text{odd}}(t)|^2$ is even, so the left-hand side reduces to

$$\begin{aligned}
\frac{1}{2}\int_0^2 |f(t)|^2 \, \mathrm{d}t &= \frac{1}{2}\int_0^1 t^2 \, \mathrm{d}t + \frac{1}{2}\int_1^2 (2-t)^2 \, \mathrm{d}t, \\
&= \frac{1}{6}\left[t^3\right]_0^1 - \frac{1}{6}\left[(2-t)^3\right]_1^2, \\
&= \frac{1}{3}.
\end{aligned}$$

Substituting this into Parseval's theorem and multiplying by $\pi^4/32$ gives

$$\sum_{k=1}^{\infty} \frac{1}{(2k-1)^4} = \frac{\pi^4}{32} \cdot \frac{1}{3} = \frac{\pi^4}{96}.$$